

Jharkhand University of Technology, Ranchi**Diploma 6th Semester Examination, 2025****Subject : Network Security****Subject Code : CSE 604****Time Allowed : 3 Hours****Full Marks : 80****Pass Marks : 26***Answer in your own words.**Answer any five questions in which Question No. 1 is compulsory.**The figure in the margin indicate full marks.**All questions carry equal marks.***1. Objective type questions and choose the right options :****2×8=16****(i) Which of the following is vulnerability at the Data Link Layer?**☒ (a) IP spoofing☐ (b) TCP reset☒ (c) ARP cache poisoning☐ (d) SYN flooding**(ii) Which encryption algorithm uses a Feistel cipher structure?**☒ (a) RSA☒ (b) AES☐ (c) DES☐ (d) ELGamal**(iii) Which of the following protocols is vulnerable to SYN flooding attacks?**☒ (a) UDP☒ (b) ICMP☐ (c) TCP☐ (d) ARP**(iv) Which of the following is an asymmetric key algorithm?**☒ (a) AES☐ (b) DES☒ (c) RSA☒ (d) 3DES**(v) Which firewall type filters traffic based on application layer data?**☒ (a) Packet-filtering firewall☐ (b) Circuit-level gateway☐ (c) Stateful inspection firewall☒ (d) Application-layer firewall**(vi) What is the purpose of DNSSEC?**☒ (a) Encrypt DNS queries☐ (b) Prevent DNS spoofing☒ (c) Enable IP spoofing☐ (d) Block malware

(vii) What is the main difference between a virus and a worm?

- (a) Virus is standalone; worm needs a host.
- (b) Worms replicate themselves; viruses don't.
- (c) Worms are faster and need no host file.
- (d) Virus spreads only through USB.

(viii) Which of the following best describes a honeypot?

- (a) A firewall type
- (b) A fake system to lure attackers
- (c) A hashing algorithm
- (d) A type of malware

2. (a) Explain the working of the Internet with a neat architecture diagram.

(b) What is an IP address? Explain its types with examples.

8+8

3. (a) Describe in detail the ARP protocol and how ARP cache poisoning attack works.

(b) Explain the TCP session hijacking attack with an example.

8+8

4. (a) Describe the working of the RSA algorithm with suitable example.

(b) Discuss about Cryptography. Explain types of Cryptography in detail.

8+8

5. (a) Explain the **process** of Playfair Cipher. What will be the cipher for the given string "Attack on Friday" with the key "APPLE" using Playfair Cipher?

(b) Explain encryption and decryption **process** in the Double DES algorithm.

8+8

6. (a) What are different types of firewalls? Explain functions with diagrams.

(b) Define IDS. How does it help in enhancing network security?

8+8

7. Write short notes on *any four* of the following:

4×4

- (a) PGP
- (b) Honeypots
- (c) DDOS attack
- (d) IP spoofing
- (e) VPN